

Euler's Basel Problem

Avery Davis

The Basel Problem, first posed in 1650, asks for the exact sum of reciprocals of squares.

For almost a century, no mathematician could calculate the sum until Euler

proved it equaled $\frac{\pi^2}{6}$ in 1734.

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = 1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \cdots = \frac{\pi^2}{6}$$

We know the series converges by integral test as

$$\int_1^{\infty} \frac{1}{n^2} dn = \lim_{R \rightarrow \infty} \left[\frac{1}{n^2} \right]_1^R = (0) - (-1) = 1.$$

Proof

From Taylor Series we know

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$$

$$\frac{\sin x}{x} = 1 - \frac{x^2}{3!} + \frac{x^4}{5!} - \frac{x^6}{7!} + \cdots$$

$$\frac{\sin x}{x} = 1 - \frac{x^2}{6} + \frac{x^4}{120} - \frac{x^6}{5040} + \cdots$$

We also know $\sin x = 0$, when $x = n\pi$

For $\frac{\sin x}{x}$, the zeroes are at

$$x = \pm\pi, \pm 2\pi, \pm 3\pi, \cdots$$

We can treat $\frac{\sin x}{x}$ as an infinite polynomial in x^2 whose roots are $x^2 = \pi^2, 4\pi^2, 9\pi^2, \cdots$

If a polynomial $P(y)$ has roots $a_1, a_2, \cdots$ and $P(0) = 1$, then $P(y) = \left(1 - \frac{y}{a_1}\right) \left(1 - \frac{y}{a_2}\right) \cdots$

Here, we can set $y = x^2$ and $a_n = n^2\pi^2$. Then $\frac{\sin x}{x} = \left(1 - \frac{x^2}{\pi^2}\right) \left(1 - \frac{x^2}{4\pi^2}\right) \left(1 - \frac{x^2}{9\pi^2}\right) \cdots$

The product is $\prod_{n=1}^{\infty} \left(1 - \frac{x^2}{n^2\pi^2}\right)$, where we see the x^2 term in $-\frac{x^2}{n^2\pi^2}$.

Therefore, the x^2 coefficient in the product is $-\sum_{n=1}^{\infty} \frac{1}{n^2\pi^2} = -\frac{1}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2}$.

From the Taylor Series, we know the coefficient of x^2 is $-\frac{1}{6}$.

Thus $-\frac{1}{6} = -\frac{1}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2}$, and multiplying by $-\pi^2$ we get $\boxed{\frac{\pi^2}{6} = \sum_{n=1}^{\infty} \frac{1}{n^2}}.$

Conceptually, this identity reveals that an arithmetic series with purely integer reciprocals is equivalent to a rational multiple of π^2 . Euler's approach treats

$\frac{\sin x}{x}$ as an infinite product over its zeroes, then expands the product into a

power series. Equating the x^2 coefficient of the product expansion with the x^2

coefficient of the Taylor series yields $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$. This demonstrates

that factorization of functions via their zeroes can give exact values with infinite series.